Geometry
Unit 1
Lesson 8 Homework

Name Answer Key__________
Date______________________
Period_____________________

Line segment WZ is shown where $WX = 2x$, $XY = 3x + 1$, $YZ = 8$, and $WZ = 29$.

1. Find the value of x.

$$2x + 3x + 1 + 8 = 29$$
$$5x + 9 = 29$$
$$5x = 20$$
$$x = 4$$

2. Find the length of $\overline{WX}$.

$$\overline{WX} = 2x$$
$$= 2 \cdot 4$$
$$= 8$$

3. Find the length of $\overline{XY}$.

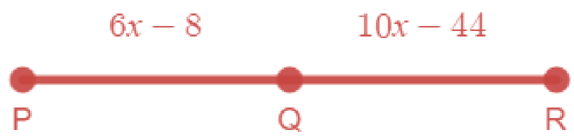
$$\overline{XY} = 3x + 1$$
$$= 3 \cdot 4 + 1$$
$$= 12 + 1$$
$$= 13$$

4. Find the length of $\overline{XZ}$.

$$\overline{XZ} = \overline{XY} + \overline{YZ}$$
$$= 13 + 8$$
$$= 21$$

P, Q, and R are three collinear points that exist such that $\overline{PQ}$ and $\overline{QR}$ are congruent. $PQ = 6x - 8$ and $QR = 10x - 44$.

5. Draw a diagram that represents points P, Q, and R.



6. Find the value of x.

$$6x - 8 = 10x - 44$$
$$36 = 4x$$
$$9 = x$$

7. How long is segment $\overline{PQ}$?

$$6x - 8 = \overline{PQ}$$
$$= 6 \cdot 9 - 8$$
$$= 46$$

8. How long is segment $\overline{QR}$?

$$10x - 44 = \overline{QR}$$
$$= 10 \cdot 9 - 44$$
$$= 46$$

9. How long is segment $\overline{PR}$?

$$\overline{PQ} + \overline{QR} = \overline{PR}$$
$$= 46 + 46$$
$$= 92$$

10. Use a compass and straightedge to make a copy of this segment.